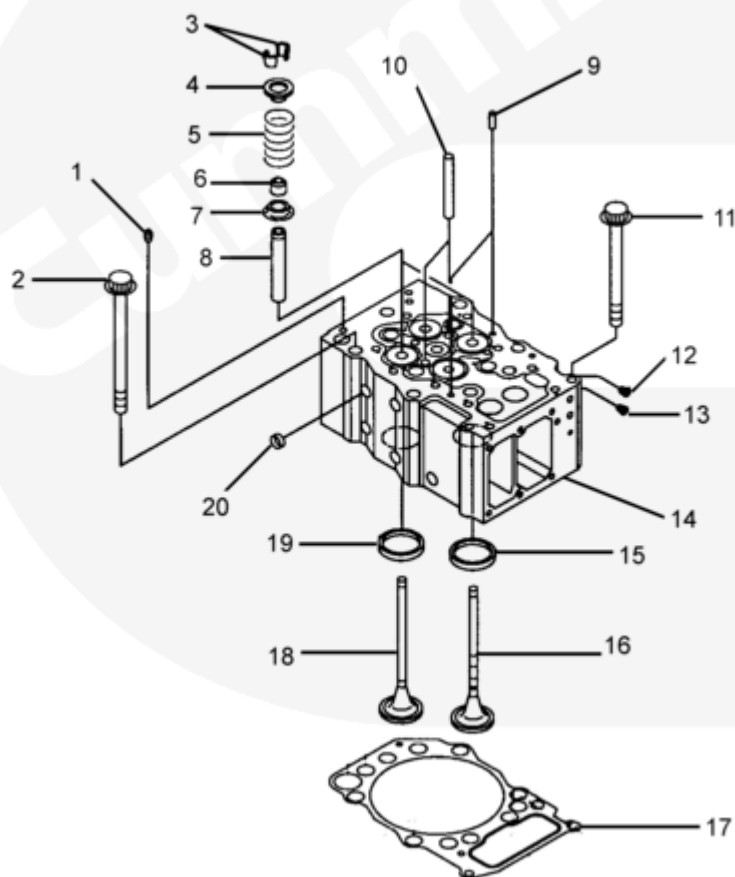


Exploded View

©Cummins Inc



02400128

1. Pipe plug
2. Capscrew, 12-point (7 large)
3. Valve collet
4. Valve spring retainer, upper
5. Valve spring
6. Valve spring seal
7. Valve spring retainer, lower

8. Valve guide
9. Dowel pin
10. Crosshead guide
11. Capscrew, 12-point (2 small)
12. Pipe plug
13. Pipe plug
14. Cylinder head
15. Exhaust valve seat
16. Exhaust valve
17. Cylinder head gasket
18. Intake valve
19. Intake valve seat
20. Expansion plug.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

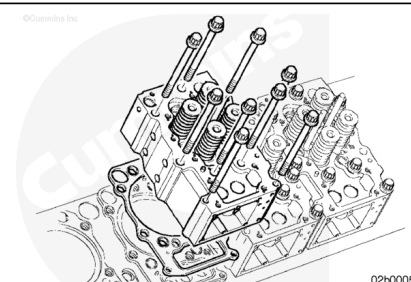
This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Disconnect the batteries or air supply to the air starter to reduce the possibility of accidental starting of the engine.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>)
- Remove the crankcase breather, if necessary. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-001-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/89/89-011-007.html>)
- Remove the intake manifold. Refer to Procedure 010-023 in Section 10. (</qs3/pubsys2/xml/en/procedures/89/89-010-023.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/89/89-004-014.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-013.html>)
- Remove the injector. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/89/89-006-026.html>)

Remove

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Remove the coolant vent line from top of the cylinder heads.

Remove the nine cylinder head mounting capscrews.

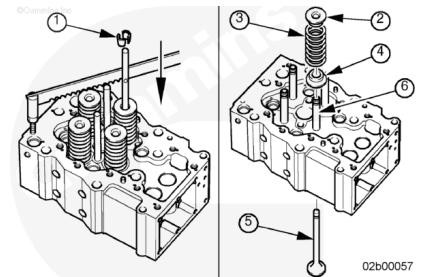
Remove the cylinder head.

Remove and discard the cylinder head gasket.

Disassemble

⚠ WARNING ⚠

The valve springs are under compression. Be cautious when using the valve spring compressor to reduce the possibility of personal injury.



Use a valve spring compressor, Part Number 3163606 or the equivalent) or a valve spring compressor stand, Part Number ST-1022 (or the equivalent). A valve spring compressor plate, Part Number 3164594 can be used with 3163606 and ST-1022 to compress all four valve springs at the same time. An air operated valve spring compressor, Part Number 3375960 (or the equivalent) can also be used.

Mark the valve for location prior to removal. The intake and exhaust valves are different. This practice will aid the assembly procedure.

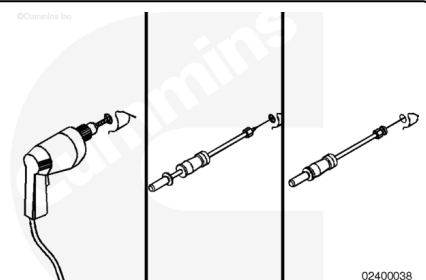
Remove the following parts:

1. Valve collet
2. Valve spring retainer
3. Valve spring
4. Wear plate
5. Valve
6. Valve stem seal.

Discard the valve collets and valve stem seals.

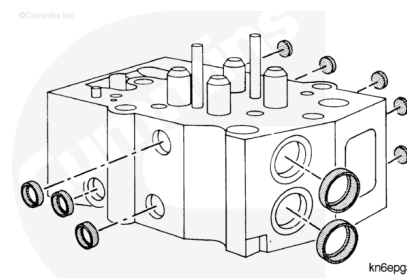
Use a drill, sheet metal screw, and the following parts from the light-duty puller kit, Part Number 3375784, to remove the cup plugs.

- Slide hammer
- Hook.



Remove and discard the 17 cup plugs.

Cup plugs **must** be removed from the head casting for cleaning purposes.

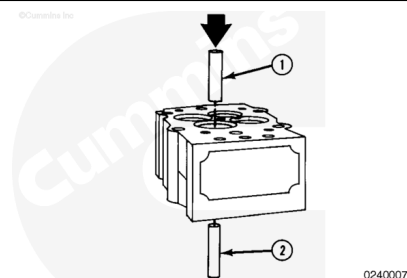


⚠ CAUTION ⚠

If a valve guide is replaced, the valve seat insert for the valve guide must be ground or severe engine damage can result.

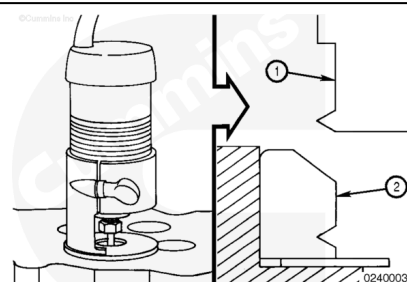
Only replace the valve guide if it is **not** within specifications.

Use an arbor press and valve guide replacer (1), Part Number 3164427, to remove the worn valve guide (2).

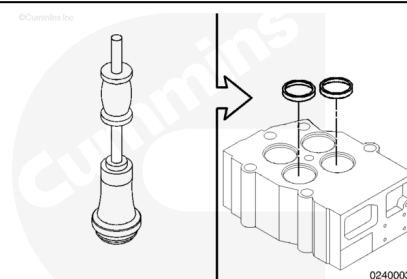


Use a valve seat cutter kit, Part Number 3376405, or equivalent.

Machine a groove in the seat (2) as close to the bottom of the bore as possible. This will allow the valve seat extractor (1) to be used.



Use a slide hammer remover, Part Number 3376617, and seat extractor, Part Number ST-1323-1 to remove the valve seats.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

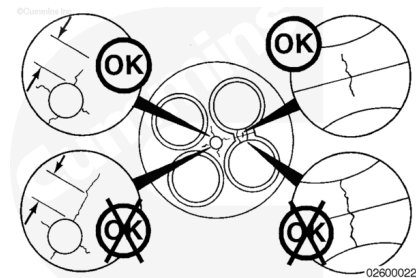
Clean the cylinder heads, capscrews, valve spring retainers, and valves with solvent.

Check for cracks on the combustion surface.

If a crack around the injector bore exceeds the maximum length, the cylinder head **must** be replaced.

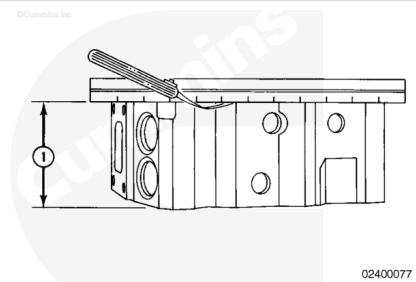
mm		in
6.4	MAX	0.250

Both ends of a crack between the valve **must** be visible. If one crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder head is questionable.



Use a straightedge to measure the flatness of the head gasket surface.

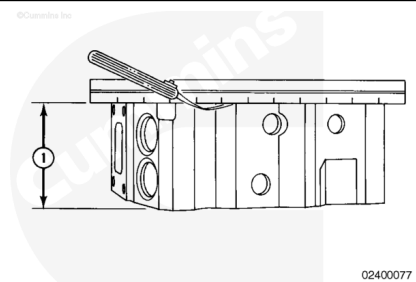
The head **must** be resurfaced, or replaced if a feeler gauge larger than 0.10 mm [0.004 in] will fit between the straightedge and the cylinder head.



The head can be resurfaced as long as the head thickness (1) measurement is within specifications.

mm		in
150.65	MIN	5.93

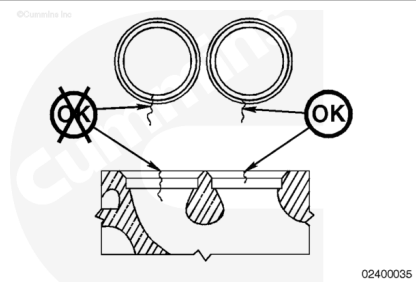
If the head is resurfaced, make certain the injector protrusion and valve depth in the head are adjusted properly.



Clean the valve seat insert bore.

Check the length of any cracks extending into the valve seat insert bore.

The cylinder head **must** be replaced if a crack extends into the bottom of the bore.



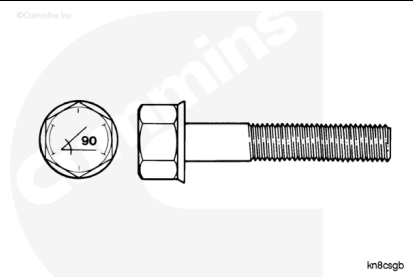
Sometimes it is possible to remove the crack by machining the head to use an oversize valve seat insert.

Note : The torque plus angle method is used only on the seven larger 233 mm [9.173 in] cylinder head capscrews.

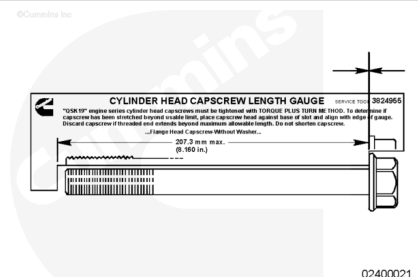
The torque plus angle method of installation places the cylinder head capscrew beyond the yield point and permanently stretches the capscrew. These capscrews can be reused throughout the life of the engine unless the capscrew exceeds the specified free length.

The free length **must** be checked to avoid bottoming in the block during installation and to make sure the capscrew still has the proper strength.

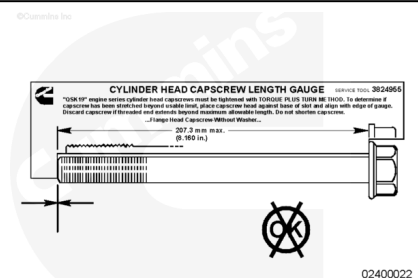
A cylinder head capscrew length gauge has been developed to check the capscrew free length.



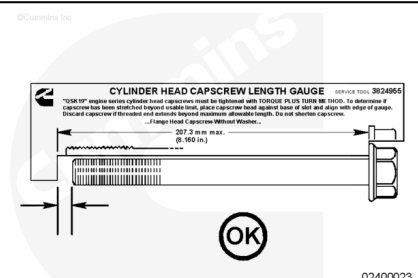
Use capscrew length gauge, Part Number 3164435, to check the capscrew free length. Place the head of the capscrew in the slot with the flange against the base of the slot.



If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.

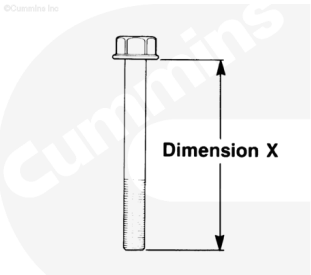


If there is clearance between the end of the capscrew and the bottom base of the tool, the capscrew is acceptable for reuse.



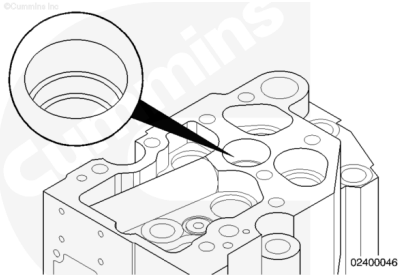
The capscrew can also be checked using a set of calipers. The maximum allowable free length is measured from the bottom of the flange to the end of the capscrew (dimension X).

mm		in
233	MAX	9.173



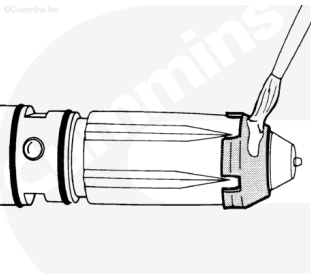
kn8csd

Inspect the injector bore for cracks or other damage.



02400046

Apply a thin coating of Prussian blue on the sealing ring.



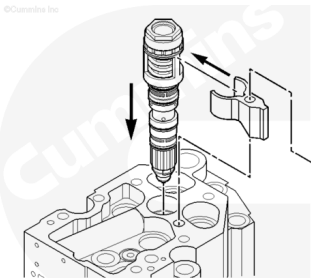
fi6sesa

⚠ CAUTION ⚠

Support the cylinder head to reduce the risk of damage to the injector tip that protrudes from the combustion face.

Install the injector and injector hold-down clamp.

Tighten the hold-down capscrew.



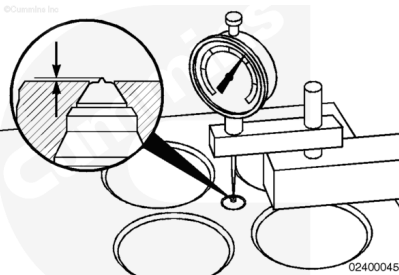
02400048

Torque Value:	Step 1	34 n.m [25 ft-lb]
	Step 2	Tighten 90 degrees.

Use a gauge block, Part Number 3164438, or equivalent, to measure the injector protrusion.

Measure as shown in the illustration.

mm		in
2.42	MIN	0.095
2.92	MAX	0.115



02400045

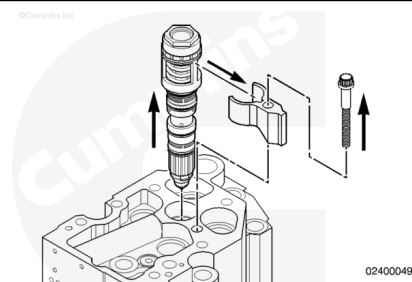
If the protrusion is **not** within specifications, use an oversize sealing ring. Sealing rings are available in various sizes.

	mm		in
Wall thickness	0.343	MIN	0.0135
	0.419	MAX	0.0165
Protrusion change	0.00	NOM	0.000
Wall thickness	0.470	MIN	0.0185
	0.546	MAX	0.0215
Protrusion change	0.25	NOM	0.010
Wall thickness	0.597	MIN	0.0235
	0.673	MAX	0.0265
Protrusion change	0.51	NOM	0.020
Wall thickness	0.724	MIN	0.0285
	0.800	MAX	0.0315
Protrusion change	0.76	NOM	0.030

The 0.343/0.419 mm [0.0135/0.0165 in] is the standard sealing ring.

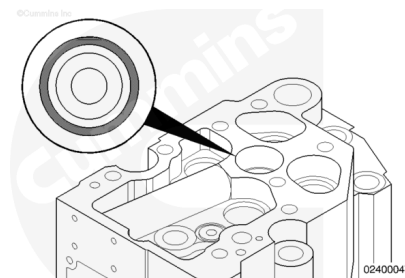
Remove the injector hold-down clamp capscrew.

Remove the injector and injector hold-down clamp.

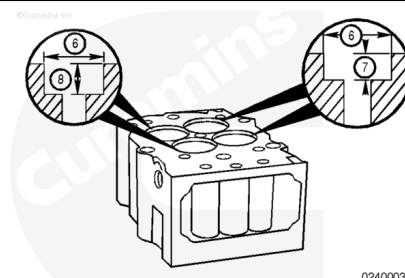


Inspect the contact pattern in the injector bore. A blue band 1.52 mm [0.060 in] wide and 360 degrees full circumference minimum **must** be visible.

If the pattern does **not** meet the specification, clean the injector bore with a tapered brush and inspect the seating pattern again. The cylinder head **must** be replaced or repaired if it is **not** within specifications.



If the valve seat inserts have been removed, measure the inside diameter and the depth of the insert bore in the head.



	mm		in
Intake Diameter	64.00	MIN	2.519
	64.019	MAX	2.520
Intake Depth	11.30	MIN	0.445
	11.50	MAX	0.452
Exhaust Diameter	63.60	MIN	2.504
	63.619	MAX	2.505
Exhaust Depth	12.20	MIN	0.480
	12.40	MAX	0.448

If the inside diameter is **not** within specifications, use an oversize seat.

Oversize valve seat inserts are available in the following sizes. Machine the valve insert bore in the head. A press fit within 0.064 to 0.114 mm [0.0025 to 0.0045 in] between the valve seat insert and the valve seat insert bore **must** be maintained.

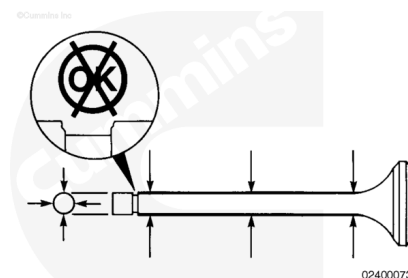
Outside Diameter Oversize	Depth (Thickness) Oversize
0.25 mm [0.010 in]	Standard
0.50 mm [0.020 in]	0.12 mm [0.005 in]
0.75 mm [0.030 in]	0.25 mm [0.010 in]
1.00 mm [0.040 in]	0.38 mm [0.015 in]

Inspect the valve collet grooves for wear.

Measure the outside diameter of the valve stem.

mm		in
11.920	MIN	0.469
11.940	MAX	0.470

mm		in
11.893	MIN	0.468
11.908	MAX	0.469



Inspect the valve springs for notches or cracks.

If the notches or cracks are present do **not** reuse the valve springs.

Test the spring force of the valve springs with service tool, Part Number 3375182.

Measurements

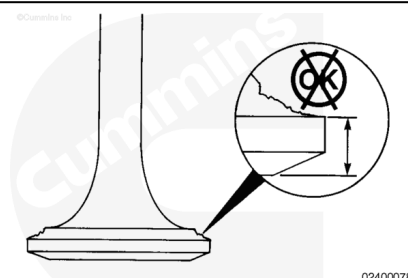
	mm	in
Approximate Free Length	99.6	3.92

Compress the valve spring to 87.4 mm [3.44 in].

n		lbf
582.6	MIN	130.97
712.0	MAX	160.06

To measure the head thickness of the valve, put the valve on a flat surface. Check the height of the outside diameter.

	mm		in
Intake	2.91	MIN	0.114
Exhaust	3.01	MIN	0.118

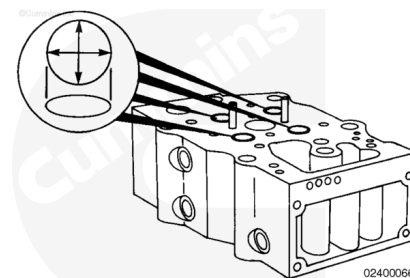


If the valve is **not** within specification, it **must** be replaced.

If the valve guides have been removed, measure the inside of the valve guide bore.

If the bore is **not** within specifications, the head **must** be replaced.

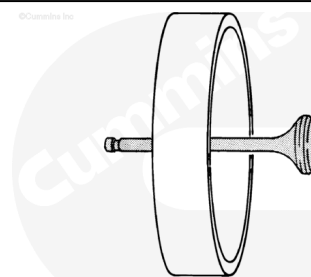
mm		in
19.00	MIN	0.748
19.021	MAX	0.749



Magnetic Crack Inspect

⚠ CAUTION ⚠

Always demagnetize and clean the parts thoroughly after a magnetic particle inspection. The iron fragments can damage the internal engine components.



Use the magnetic particle residual method.

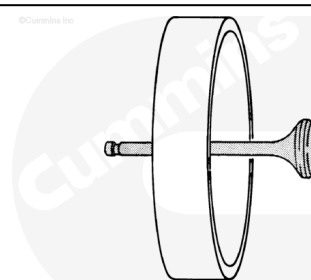
Check the valves for cracks.

Check the intake valves with the coil and head shot methods (the coil shot first). Check the exhaust valves with the coil shot method.

Use a 305-mm [12-in] minimum diameter coil.

Coil Shot Amperage (Ampere Turns)

Minimum	Maximum
400 VDC	800 VDC

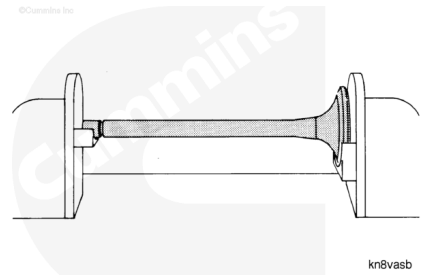


An ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

A broad, fuzzy pattern will appear at the welded joint on the exhaust valves. This is normal. If there is a distinct line in the broad, fuzzy pattern, the valve **must** be replaced.

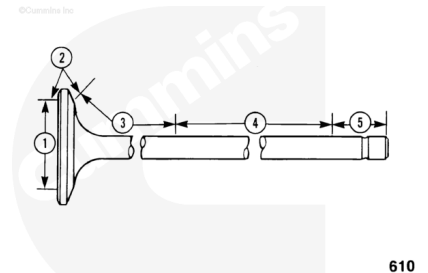
Head Shot Amperage (Ampere)

Minimum	Maximum
500 VDC	700 VDC



Acceptance criteria (intake and exhaust valves):

1. Indications less than 38.1 mm [1.5 in] in length are acceptable. More than five indications, spaced closer than 3 mm [0.118 in], are **not** acceptable.
2. No indications are acceptable.
3. **Only** longitudinal indications are acceptable.
4. No indications are acceptable.
5. No indications are acceptable.



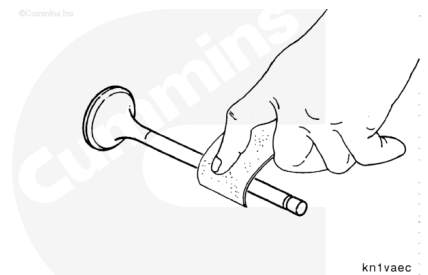
Grind

Valves

Do **not** use aluminum oxide cloth or sandpaper to clean the valves.

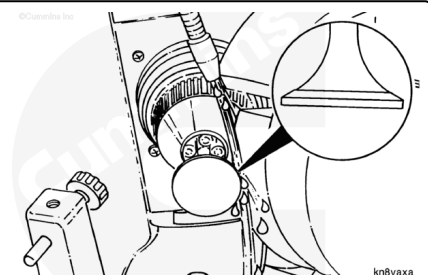
Use Scotch-Brite™, Part Number 3823258, to clean the valve stems. Clean the carbon deposits from the valve face and the head.

The valves **must** be clean and free of carbon deposits before they are ground. Valves can be cleaned by the bead blasting method in the head area **only**. If the stems are bead blasted, the valve **must not** be reused because surface finish will be damaged.



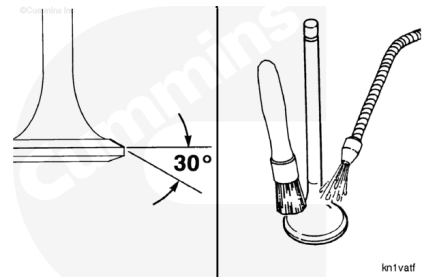
Use a valve facing machine, Part Number 3376256 or equivalent, to grind the face of the valve.

Follow the instructions supplied with the valve facing machine for the correct setup before grinding the valves.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

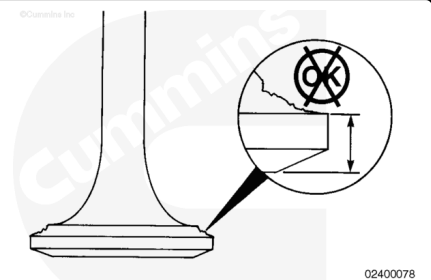
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the metal particles from the valve.

Dry with compressed air.

To measure the head thickness of the valve, put the valve on a flat surface. Check the height if the outside diameter.

	mm		in
Intake	2.91	MIN	0.114
Exhaust	3.01	MIN	0.118



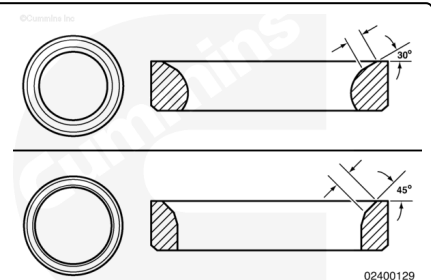
If the valve is **not** within specification, it **must** be replaced.

Valve Seat

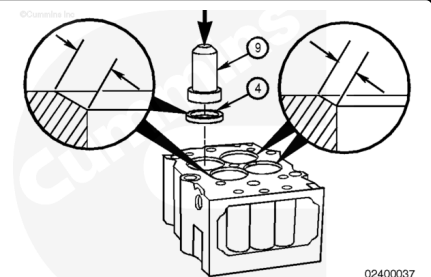
Identify the correct valve seat insert.

- Top illustration: Intake port seat.
- Bottom illustration: Exhaust port seat.

Both the intake and the exhaust port seats **must** be installed with the angle positioned as shown.



Use an arbor press with valve seat adapter insert installer, Part Number 3164790 and valve seat installer, Part Number 3164978 (9) to install the insert (4) into the head.

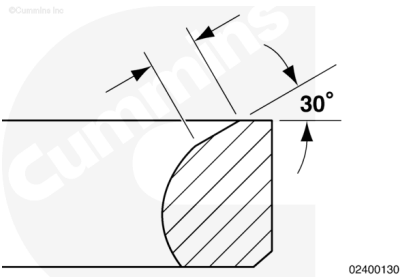


Use the valve seat grinding machine, Part Number ST-685-A or ST-685-C or equivalent, and the valve guide arbor set, Part Number 3164734 or equivalent, to grind the valve seats.

Grind the valve seat insert to the angle shown.

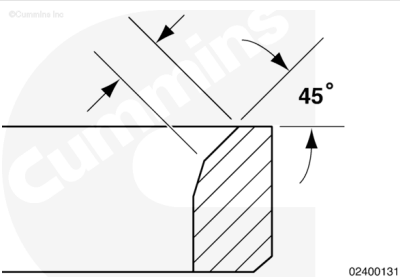
Measure the width.

	mm		in
Intake	3.5	MIN	0.138
	4.2	MAX	0.165
Exhaust	4.0	MIN	0.157
	4.3	MAX	0.169



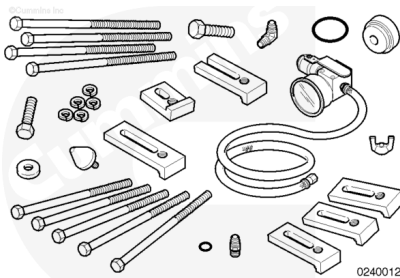
If the width of the valve seat is **not** within specifications, remove the surface material on the inside diameter and the outside diameter to decrease the width of the seat.

If the valve seat specifications are **not** obtained by grinding, the insert **must** be replaced.

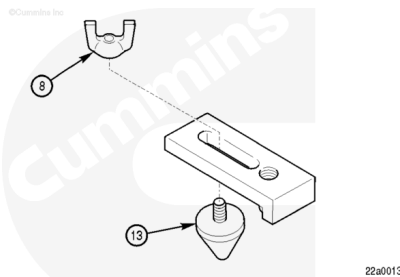


Leak Test

Use cylinder head leak test kit, Part Number 3164341, with regulator Part Number 3164231, to test the cylinder head.



Assemble the wing nut (8), clamp and plug (13) as necessary.



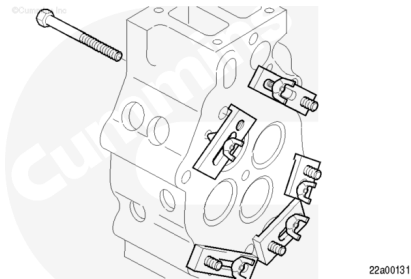
Note : The cylinder head as shown uses five clamps and plug assemblies; the QSK23 cylinder heads use **only four longer assemblies**.

Install the clamp and plug assembly into the cylinder head, blocking the coolant passages.

Insert the appropriate capscrew through the valve guide side of the cylinder head, and thread it into the clamp.

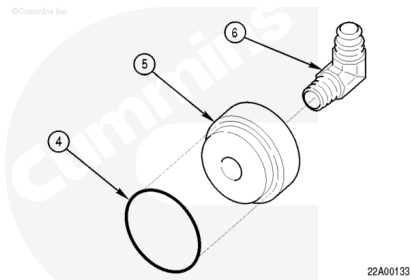
Tighten the capscrew.

Torque Value: 20 n•m [177 in-lb]



Install the o-ring (4) into the o-ring groove of the test adapter (5).

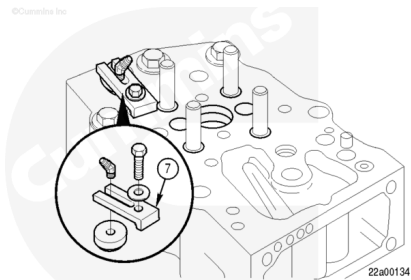
Thread the adapter elbow (6) into adapter (5).



Place the test adapter assembly into the cylinder head.

Install the clamp (7) over the test adapter assembly secure it with the appropriate capscrew and washer.

Torque Value: 54 n•m [40 ft-lb]

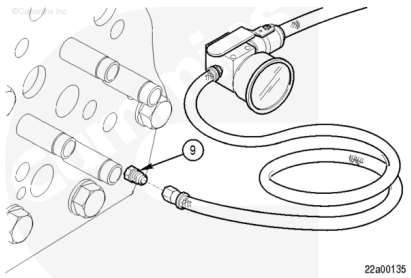


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Connect the pressure regulator valve kit to the pipe nipple (9).

Connect the pressure regulator valve kit to shop air and adjust air pressure.

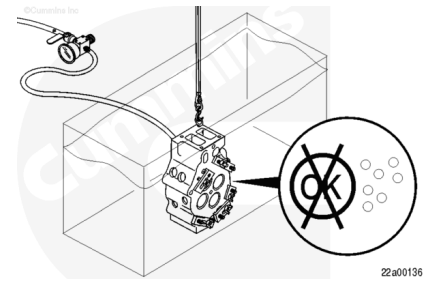


Measurements

	kpa	psi
Air Pressure	345	50

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Attach an eyebolt to the cylinder head.

Use a suitable lifting device to lift the cylinder head.

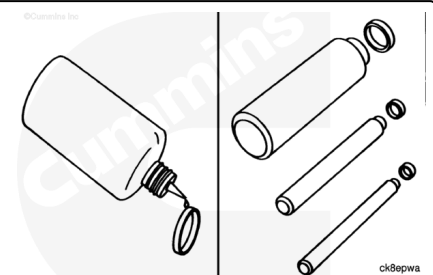
Immerse the cylinder head into a tank of water.

Check for bubbles. Bubbles indicate an air leak.

If the cylinder head leaks, it **must** be repaired or replaced.

Assemble

Use cup plug Loctite™ sealant, Part Number 3375068, or equivalent.

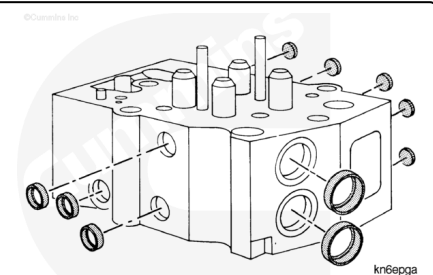


Plug O.D. 22.63 mm [0.891 in], I.D. 21.18 mm [0.834 in].

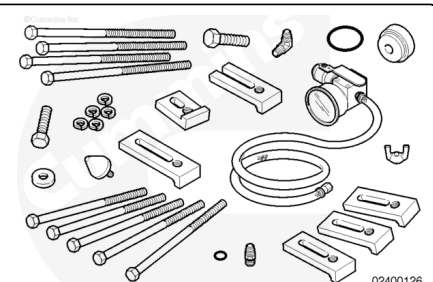
Use the correct expansion cup plug installer (driver) to make sure the expansion cup plugs are installed to the correct depth in the cylinder head. Drive the plug until the shoulder of the installer (driver) contacts the cylinder head.

Expansion (cup) plug installer handle, Part Number 3376795 or 3164085 **must** be used with expansion (cup) plug installer (driver), Part Number 3376813.

Install 17 new cup plugs.

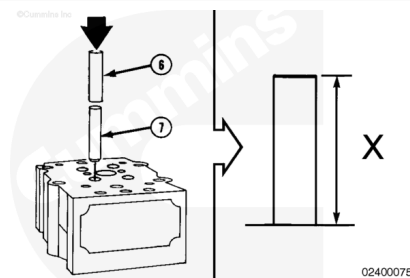


Use cylinder head leak test kit, Part Number 3164341, with regulator, Part Number 3164231, to test the cylinder head.



⚠ CAUTION ⚠

Make certain the mandrel does not damage the top of the valve guide. If the guide is damaged, the valves may stick, causing severe engine damage.



If valve guides were removed, use an arbor press and a suitable mandrel, Part Number 3164427, to install the new valve guides.

Install the valve guide (7) to the specified height (X).

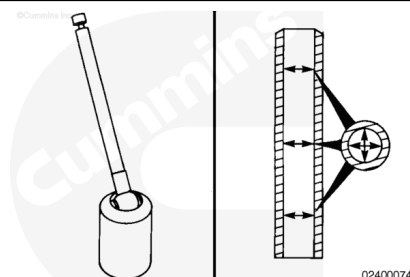
	mm		in
To top of head	42.50	MIN	1.673
	43.50	MAX	1.713

Inspect the valve guides for chips and cracks.

Use a ball gauge or a dial bore indicator to measure the inside diameter of the valve guides.

Measure the inside diameter.

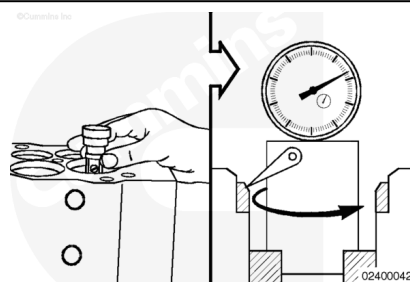
mm		in
11.981	MIN	0.471
11.999	MAX	0.472



Measure the valve seat to the valve guide concentricity.

The seat and guide be concentric.

mm		in
0.05	MAX	0.002



Use a lead pencil or Prussian blue to mark across the valve face. Install the valve in the valve guide.

Hold the valve against the valve seat and rotate the valve backward and forward three or four times.

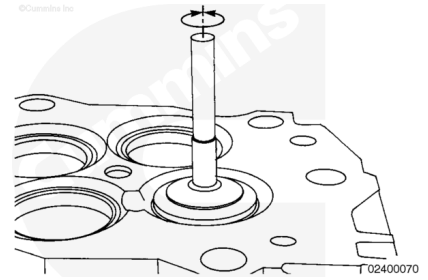
Correct contact against the valve seat will break the marks on the valve face.

Valves and valve seats that are correctly machined do **not** require the use of lapping compound to make an airtight seal. If lapping compound is required, inspect the adjustments of the facing machine and the condition of the grinding stone.

Use a fine lapping compound, Part Number 3375805, or equivalent. Apply a thin and even coating on the valve.

Use a power or a hand suction lapping tool to provide pressure in the center of the valve.

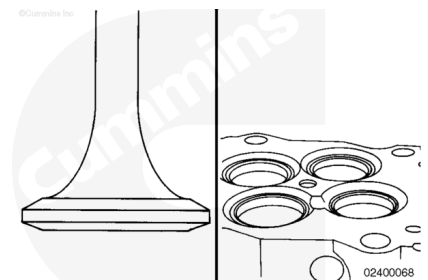
Turn the valve backward and forward. Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.



⚠ CAUTION ⚠

Lapping compound is an abrasive material. Failure will result if the cylinder head, valves, and valve seats are not cleaned thoroughly.

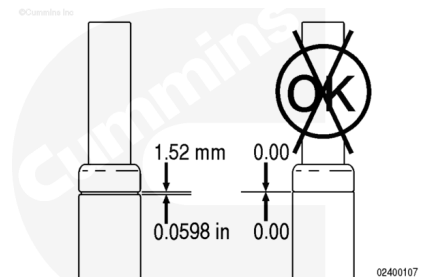
Clean the lapping compound from the parts.



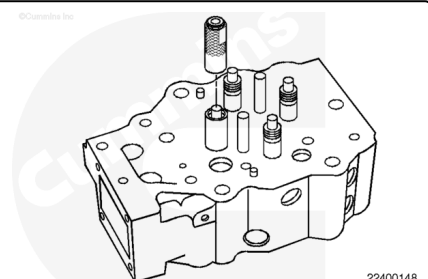
⚠ CAUTION ⚠

To avoid the valve seal fitting loosely on the valve guide, do not lubricate the valve seal. The seal must be installed dry.

If the valve stem seal is installed too far on the guide, the seal will fail prematurely, causing excessive oil consumption. The gauge allows correct positioning of the seal.

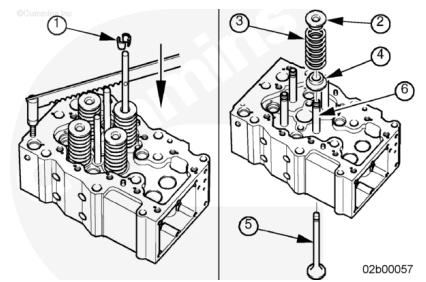


Use valve stem seal installer, Part Number 3164430, to install the valve stem seals.



⚠ CAUTION ⚠

The valve springs are under compression. Use caution when using the valve spring compressor. Personal injury can result.



Use a valve spring compressor, Part Number 3163606 (or equivalent), or a valve spring compressor stand, Part Number ST-1022 (or the equivalent). A valve spring compressor plate, Part Number 3164594 can be used with 3163606 and ST-1022 to compress all four valve springs at the same time. An air operated valve spring compressor, Part Number 3375960 (or the equivalent) can also be used.

Assemble the parts in the following order:

- Valve (5)
- Wear plate (4)
- Valve spring (3)
- Valve spring retainer (2).

Compress the valve spring and install the following parts:

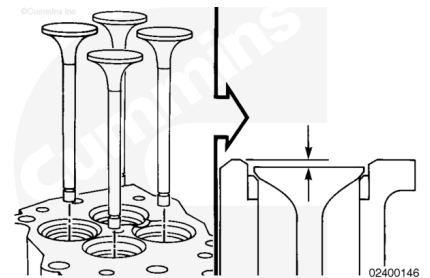
- New valve collets.

Install the valve into the valve guide.

Measure the depth of the valve with a depth gauge, Part Number 3164438, or equivalent.

The valve head **must** be even with or **not** more than 0.80 mm [0.031 in] below the surface of the cylinder head.

If the valve depth is **not** within specifications, the valve or the valve seat **must** be replaced.

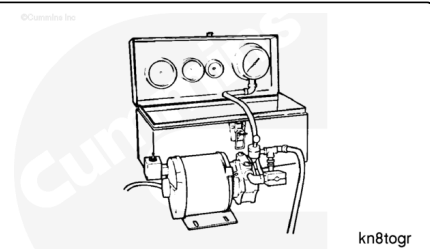


Vacuum Test

Use valve vacuum tester, Part Number 3824277 for 115 VAC, or Part Number 3824278 for 220 V, with vacuum cup, Part Number 4918162, and vacuum cup seal, Part Number 4918163 to test valves

Before using the tester, test the leakage shutoff valve. If the valve is dirty or worn, it will produce a false leakage measurement on the gauge.

Check the valve:

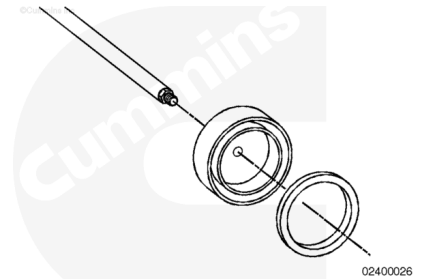


- Open the shutoff valve.
- Turn on the vacuum pump.
- Place the cup against a smooth surface.
- Close the shutoff valve.
- Turn off the vacuum pump.
- Wait approximately 10 seconds.
- The gauge **must not** drop more than 7 kPa [2 in Hg].

The valve and the valve seats **must** be clean and dry.

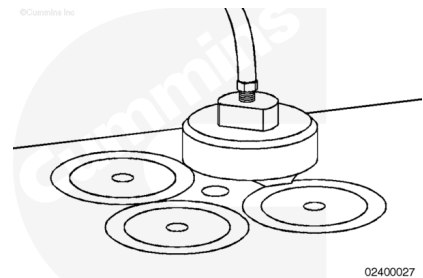
Choose the correct cup for the engine model that is to be tested.

Install the seal and the cup to the vacuum line (hose).



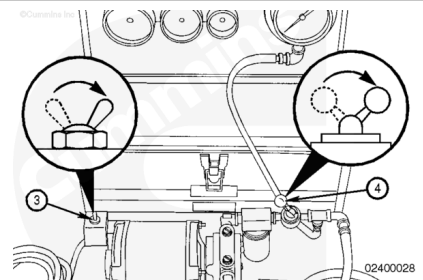
Cover the valve with the cup and the seal. The seal **must** have a tight contact on the cylinder head around the valve.

To check the valves, the seal **must** completely fill the milled area between the valves.



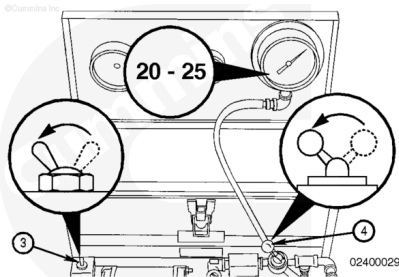
Move the toggle switch (3) to the ON position.

Turn the vacuum control valve (4) to the OPEN position.



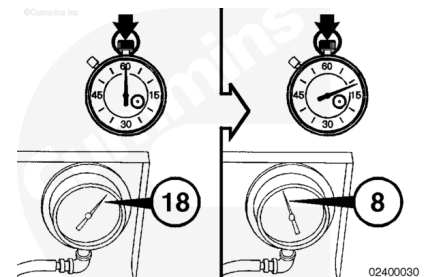
When the gauge indicates between 10 in Hg and 25 in Hg, turn the vacuum control valve (4) to the CLOSED or OFF position.

Turn the toggle switch (3) to the OFF position.



⚠ CAUTION ⚠

The cylinder head must be disassembled and cleaned after any grinding or cutting procedures to reduce the possibility of engine damage.



Use a stopwatch. As the vacuum gauge needle moves **counterclockwise**, start the timing when the needle on the gauge points to 18 in Hg.

Stop the timing when the needle on the gauge points to 8 in Hg.

The elapsed time for the gauge to move between the specified readings **must** be 10 seconds or more.

If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be certain the equipment is functioning properly.
- Use a mallet to hit the valve stem lightly to be certain the valve is sealing. Repeat the test.
- Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.
- If the leakage is between the insert and the head, the insert **must** be replaced.

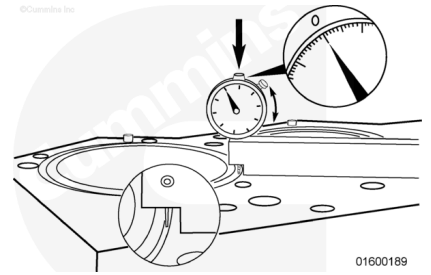
If the leakage is between the valve and the insert, one of the following procedures **must** be performed:

- Lap valve to seat
- Grind the valves
- Grind the valve seat.

Install

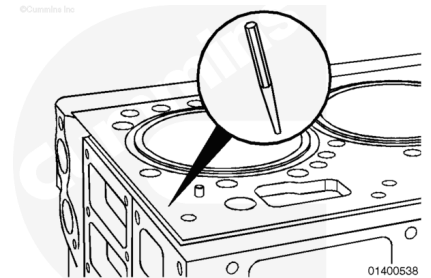
Measure the liner protrusion. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-028.html>)

If the liner protrusion is **not** correct, it **must** be adjusted before installing the cylinder head.

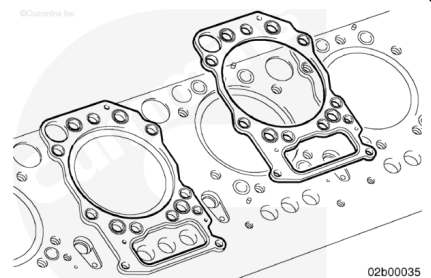


An oversize cylinder head gasket is often required on cylinder blocks that have had material removed from the top surface of the cylinder block.

Inspect the cylinder block for markings to determine if a oversize or a standard size cylinder head gasket is required.

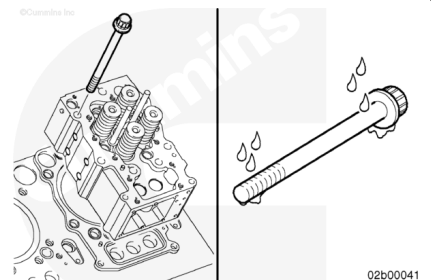


Install the cylinder head gasket, using dowel pins for alignment.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



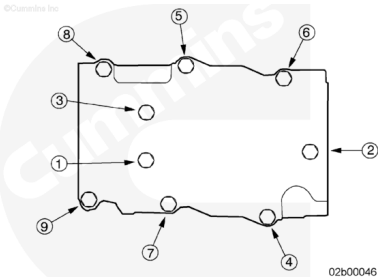
Install the cylinder head. It **must** slide easily over the dowel pins.

Lubricate the capscrew cylinder head flange with clean engine oil.

Lubricate the capscrew threads with clean engine oil. Allow excess oil to drip off the capscrews before installing them into the block.

Install the capscrews.

When using torque plus angle, the tolerance on the 90 degree angle of rotation is 90 degrees $\pm$ 5 degrees. If the capscrew is rotated beyond two flats, do **not** loosen the capscrew. The clamp load is still acceptable; however, rotating the capscrew beyond two flats causes additional stretch and reduces the life of the capscrew. With proper tightening, the capscrew can typically be reused for the life of the engine.



Tighten the capscrews 1 through 7 in sequence.

Torque Value:	Step 1	100 n.m [74 ft-lb]
	Step 2	245 n.m [181 ft-lb]
	Step 3	380 n.m [280 ft-lb]
	Step 4	Rotate 90 degrees

Tighten capscrews 8 and 9 in sequence.

Torque Value:	Step 1	50 n.m [37 ft-lb]
	Step 2	100 n.m [74 ft-lb]

Replace the coolant vent line.

Torque Value: 12 n•m [106 in-lb]

Finishing Steps

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-013.html)

- Install the injector. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/89/89-006-026.html>)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/89/89-004-014.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-009.html>)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-006.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/89/89-003-011.html>)
- Install the intake manifold and crossover. Refer to Procedure 010-023 in Section 10. (</qs3/pubsys2/xml/en/procedures/89/89-010-023.html>)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/89/89-011-007.html>)
- Connect the batteries or air starter supply line.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>)
- Operate the engine to 70°C [158°F] minimum coolant temperature and check for leaks.